Designing Decision-Centric Banking Analytics Systems:

A Practical Framework from Regional Banking Environments

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Abstract

Analytics systems in financial institutions often fail not due to lack of data, but due to their inability to influence operational decision-making. This challenge is particularly pronounced in regional banking environments, where data is fragmented across multiple sources, reporting processes are manual, and user adoption of analytics tools remains low.

This paper presents a practical framework for designing decision-centric analytics systems in low data-maturity environments. Based on a real-world implementation across 27 banking branches, the study outlines a three-layer model consisting of data standardisation, decision-focused dashboard design, and user adoption alignment.

The implementation resulted in a 40% reduction in reporting time and a 15% improvement in non-performing asset (NPA) recovery performance. The findings highlight that the effectiveness of analytics systems depends less on technical sophistication and more on their integration into operational workflows and decision-making processes.

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1. Introduction

The increasing availability of data in financial institutions has led to widespread adoption of analytics tools. However, in many regional and mid-sized banking environments, the presence of data does not translate into effective decision-making.

A key limitation lies in the design of analytics systems, which often prioritise data aggregation and visualisation over usability and operational integration. As a result, reporting processes remain manual, insights are delayed, and decision-making continues to rely on fragmented information.

This paper explores how analytics systems can be designed to improve decision effectiveness, using a real-world case from regional banking operations.

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2. Industry Challenges

Regional banking environments face several structural challenges:

- Data fragmentation across core banking systems, manual registers, and customer records
- Time-intensive and backward-looking reporting processes
- Lack of unified visibility into portfolio performance
- Low adoption of analytics tools among operational staff

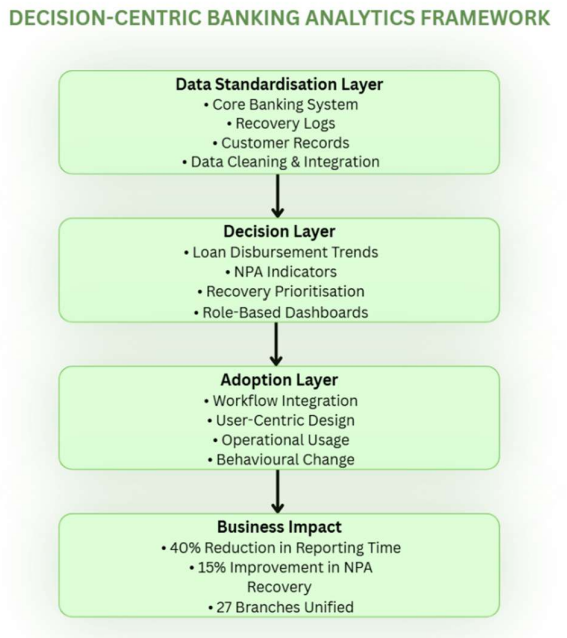
These challenges create a disconnect between data availability and actionable insights, limiting the effectiveness of analytics initiatives.

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3. Methodology: A Three-Layer Analytics Framework

To address these challenges, a three-layer framework was developed for designing decision-centric analytics systems.

Figure 1: Decision-Centric Banking Analytics Framework



3.1 Data Standardisation Layer

The foundation of the framework involves consolidating fragmented data sources into a unified structure.

Key activities include:

- Integration of data from core banking systems, recovery logs, and customer records
- Application of consistent classification logic aligned with regulatory categories (e.g., NPA ageing buckets)
- Data cleaning and standardisation

This layer ensures that all subsequent analysis is based on reliable and consistent data.

3.2 Decision Layer

The second layer focuses on designing analytics outputs that directly support operational decisions.

Instead of presenting comprehensive datasets, dashboards are structured around specific questions such as:

- Which accounts are most at risk of default?
- Where should recovery efforts be prioritised?

Role-based dashboards were developed for different stakeholders, including branch leadership, recovery teams, and regional management.

3.3 Adoption Layer

The final layer addresses user adoption, which is often the primary barrier to analytics success.

Key considerations include:

- Alignment of dashboards with daily workflows
- Focus on simplicity and clarity
- Co-design with end users based on their decision-making needs

This layer ensures that analytics tools are not only available but actively used.

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4. Implementation

Figure 2: Operational Implementation of Loan Portfolio Analytics System

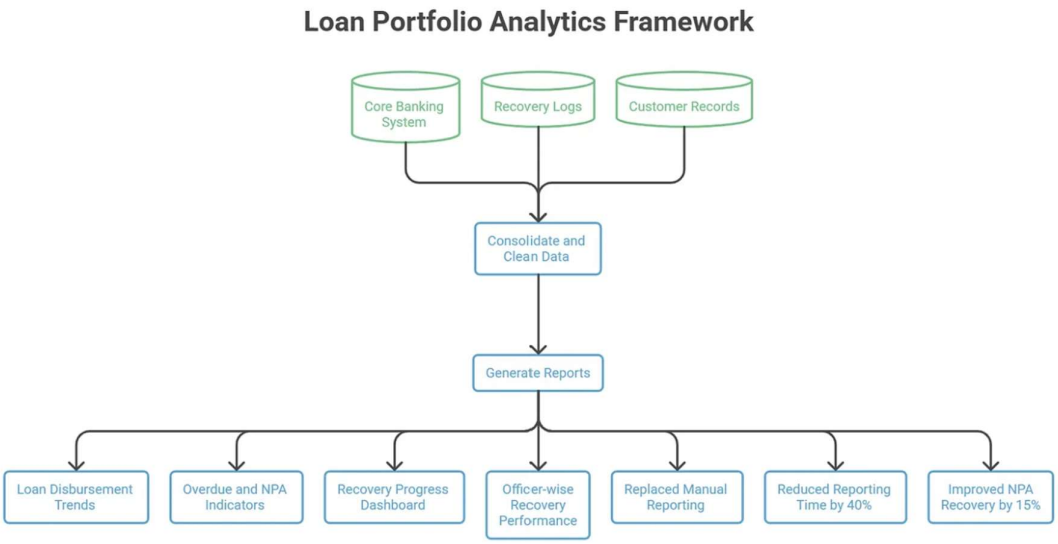


Figure 2 illustrates the data flow and reporting structure implemented across multiple branches, integrating core banking systems, recovery logs, and customer records into decision-support dashboards.

The framework was implemented in a regional banking environment covering:

- 27 branches
- Thousands of active loan accounts
- Multiple data sources including core banking systems and manual records

Technology Stack:

- Microsoft Excel for data modelling, integration, and classification
- Power BI for dashboarding and reporting

The system replaced manual reporting processes with automated, standardised analytics workflows.

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5. Results

The implementation led to measurable improvements:

- **40% reduction in monthly reporting time**
- **15% improvement in NPA recovery performance**
- **Unified reporting across 27 branches**

In addition to efficiency gains, the system enabled a shift from reactive to proactive decision-making.

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6. Discussion

The findings from this implementation highlight several key insights:

6.1 Data Quality is Organisational

Data issues are often driven by inconsistent practices rather than technical limitations. Establishing standard definitions and processes is critical.

6.2 Simplicity Drives Adoption

Dashboards that address a single operational question are more effective than complex, multi-purpose reports.

6.3 Analytics Value Lies in Behaviour Change

The impact of analytics systems is realised when they influence how decisions are made, not when they generate insights alone.

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7. Broader Applicability

The framework is applicable beyond banking, particularly in environments where:

- Data is fragmented
- Users are non-technical
- Decisions are operational

Relevant domains include:

- Insurance operations

- Collections and recovery systems
- Government reporting frameworks
- Early-stage financial technology platforms

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8. Conclusion

The effectiveness of analytics systems should be evaluated based on their ability to support decision-making and drive operational outcomes.

In low data-maturity environments, success depends on designing systems that integrate data, decisions, and user behaviour. The proposed three-layer framework provides a practical approach for achieving this integration.